Viking MotorSports VCU Module

Fabrication and Assembly Information

For build 2026-08-13T10-19-09

Contents

About this Board

Board Description

Describe board here

Documentation Links

Link to documentation here

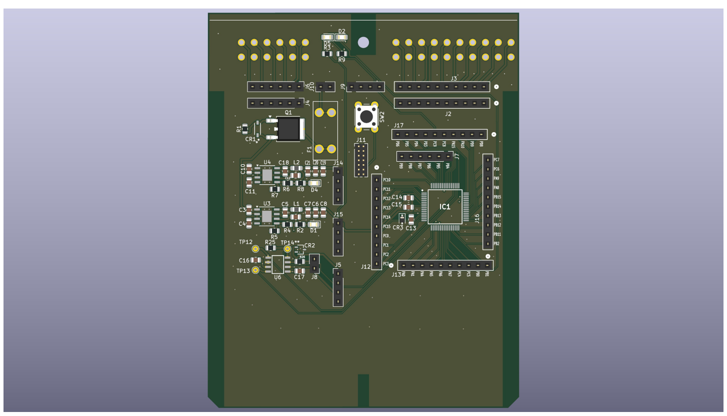
Documentation Files

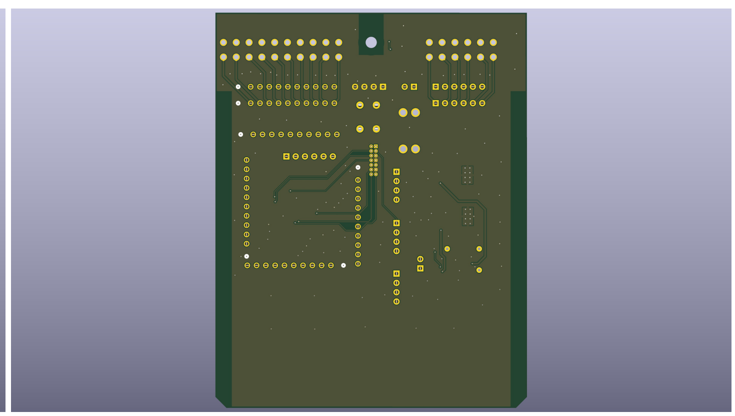
Filename	Notes
README.pdf	This README file
VCU_Module-outline.dxf	Board outline (with holes) in DXF format
VCU_Module-pcba.step	3D model of PCBA (with components)
VCU_Module-render-bot.jpg	Render of the top of the 3D model
VCU_Module-render-bot.jpg	Render of the bottom of the 3D model
VCU_Module-schematic.pdf	PDF of board schematics

Contact Information

- Website: *website*
- Email: *email*

Board Renders





Printed Circuit Board (PCB) Fabrication Information

Board Info

- 2 layer board
- Bounding box is *TBD*
- Board thickness is 1.59 mm (0.063 inch)

Board Requirements

- Design Rules
  - Minimum Trace / Space design rulea: 0.1524 mm (6.0 mil) / 0.1524 mm (6.0 mil)
  - Outer dimension router tolerance: +/- 0.254 mm (10.0 mil)
  - Hole placement tolerance: +/- 0.075 mm (3.0 mil)
- Drills
  - Drill Positional Tolerance: 0.051 mm (2.0 mil)
  - Drill Size tolerance: +/- 0.064 mm (2.5 mil)
- Plated/Un-plated holes
  - There are no un-plated (NPT) holes
  - There are 96 plated through (PTH) holes
  - PTH minimum diameter: 0.254 mm (10 mil)
  - PTH minimum annulus: 0.102 mm (4 mil) radius
- Outline/Routing
  - No requirements.
- Slots
  - There are no slots.
- Cutouts
  - There are no cutouts
- There are 3 fiducials on the top layer.
- Panel tabs ("mouse bites")
  - Card edges must be smooth; no mouse bites or other intrusions into the card outline.
  - If external mouse bites are required, minimize and customer will remove by hand before assembly.
- If not otherwise specified, build to IPC 6012 Class 2 or better.

Materials

- No requirements for materials.
- Copper Surface treatment should be ENIG, althogh immersion Silver is acceptable.
- White silkscreen on top and bottom surface
- Taiyo PSR-4000 or equivalent soldermask on top and bottom, no requirements for color.

Stack Up

- Any FR4 like material is acceptable.
- 1 oz Copper on top and bottom layers
- 1.6 mm thick when done (1/16 inch)

Array / Panel Information

- Coordinate with Contract Manufacturer (CM) for optimal size of this panel.
- If no feedback from CM, then produce single boards (no panel).

Fabrication Files

IPC-2581 File

Filename	Notes
VCU_Module-ipc2581.xml	IPC-2581 board information file

Legacy PCB Files

Filename	Notes
VCU_Module-Edge_Cuts.gbr	RS274X file for the dimension (outline) layer
VCU_Module-F_Silkscreen.gbr	RS274X file for the top silkscreen
VCU_Module-F_Mask.gbr	RS274X file for the top soldermask
VCU_Module-F_Cu.gbr	RS274X file for the top copper layer
VCU_Module-In1_Cu.gbr	RS274X file for the layer 2 copper
VCU_Module-In2_Cu.gbr	RS274X file for the layer 3 copper
VCU_Module-B_Cu.gbr	RS274X file for the bottom copper layer
VCU_Module-B_Mask.gbr	RS274X file for the bottom soldermask
VCU_Module-B_Silkscreen.gbr	RS274X file for the bottom silkscreen
VCU_Module-NPTH.drl	Excellon file for non-plated through holes
VCU_Module-PTH.drl	Excellon file for plated through holes

Printed Circuit Board Assembly (PCBA) Information

Assembly Info

- All components are on the top side of the board.
- This PCBA has two components that are SMT components with through-hole mechanical leads.

Assembly Requirements

- Assemble to IPC Class 2 or better
- Solder paste can be ROHS or leaded.
- No clean, or Aqueous flux and wash, can be used.
- No conformal coating.

Component Specific Assembly Information

- None

Assembly Files

IPC-2581 File

Filename	Notes
VCU_Module-ipc2581.xml	IPC-2581 board information file

Bill of Materials (BOM)

Filename	Description
VCU_Module-bom.csv	BOM in Comma Separated Variable format

Solder Paste Stencils

Filename	Notes
VCU_Module-B_Paste.gbr	RS274X file for top/front solder paste stencil
VCU_Module-F_Paste.gbr	RS274X file for bottom/back solder paste stencil

Mounting/Placement Location

Filename	Description
VCU_Module-3u.pos	Pick and place locations for components